



CASE STUDY

AI in Educational Publishing and Retail Supply Chain

Innovacio unified POS, ERP, supplier, and retailer data into one AI dashboard to manage a 2,000-title catalog across 4,000 outlets, addressing 4–8 month lead times, 17.3% consignment exposure, and over \$15M in noncurrent inventory risk while improving forecasting, replenishment, and margin control.

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Problem

The client's educational publishing and retail supply chain was under pressure from a large, fragmented title catalog, long overseas lead times, seasonal demand spikes, retailer complexity, and growing noncurrent inventory exposure. Buying, replenishment, and inventory decisions were being made across disconnected systems and spreadsheets, which made it difficult to know early enough what to print, where to place stock, and how much inventory to hold.

Demand varied sharply by title, channel, retailer, and season, but the planning process did not have a reliable way to separate evergreen products from seasonal or promotional items. Supplier minimums and long transit times increased the risk of overbuying, while consignment stock made it harder to see what had sold, what remained on hand, and what needed to be replenished or pulled back. As a result, the business faced stockouts during peak periods on one side and slow-moving, noncurrent inventory on the other.

How we solved it

Innovacio Technologies designed an AI-powered operating layer that connected POS, ERP, warehouse, supplier, and retail-partner data into one live decision environment. This gave the client a clearer view of demand, inventory aging, supplier lead times, and retailer replenishment needs across the network.

We implemented title- and channel-level demand forecasting to improve planning for seasonal peaks, evergreen titles, and promotional windows. We also added supplier lead-time and purchase-order intelligence so procurement could make better decisions on when to buy, how much to buy, and which titles were most at risk of missing the selling window.

To improve inventory control, we built aging and noncurrent-stock alerts that flagged titles before they became slow-moving or obsolete. We also introduced retailer replenishment and consignment visibility so the business could track partner sell-through, stock on hand, and next-best-order timing more accurately across outlets.

The result was a more disciplined retail supply chain where forecasting, procurement, replenishment, and inventory control were connected in one system, helping the client reduce emergency buys, improve retailer service levels, and protect margin from aging stock.



Executive Summary

The client operates a hybrid educational publishing and retail-distribution model with roughly 2,000 titles, a network of about 4,000 retail outlets, and a product mix that includes children's books, toys, and learning products. The company's annual report shows that most titles are printed overseas, often requiring four to eight months of lead time, while sales are strongest in the fall holiday period and again in spring around Easter. That combination creates a classic AI opportunity: demand is seasonal, supply is slow, inventory can age into noncurrent stock, and the retailer network needs accurate replenishment decisions long before demand becomes visible.

Innovacio's solution would not be a single forecast model. It would be a connected supply-chain control layer: demand sensing at SKU and channel level, procurement planning with lead-time risk scoring, retailer replenishment dashboards, consignment visibility for field partners, and inventory-aging alerts that flag titles before they become slow-moving or obsolete. In a business where noncurrent inventory is already measured in millions of dollars and supplier minimums can force overbuying, AI becomes the decision engine that links sales, buying, warehouse planning, and retail execution.

The practical outcome is a more disciplined operating model: fewer emergency buys, fewer stockouts during peak season, lower noncurrent inventory exposure, better allocation to the strongest retailers, and a clearer view of which titles and product lines deserve reprint, markdown, or retirement.

Why this business is different

- The product catalog is broad and fragmented. Roughly 2,000 titles create many demand patterns, from evergreen bestsellers to slow-moving specialty titles.
- Supply lead times are long. The company's annual report states that most products are printed in China, Europe, Singapore, India, Malaysia, and Dubai, typically with four- to eight-month lead times.
- Demand is strongly seasonal. Sales peak in the fall holiday season and rise again in spring around Easter.



- Inventory can become noncurrent quickly when minimum order quantities force overbuying or when retail demand softens.
- Consignment inventory adds operational complexity because field partners hold stock, but the company still needs visibility into sell-through, returns, and replenishment timing.
- Retail distribution is fragmented, with sales to bookstores, toy and specialty stores, museums, and other outlets that need different product mixes and service levels.

Seasonal demand forecasting for titles, toys, and gift products

Business problem: The hardest operational question is not how to sell existing inventory; it is how much to print or buy before the season starts. The client sells into retail outlets where fall holiday demand and spring Easter demand are both important. That means buying too early can lock cash into slow-moving stock, while buying too late can miss the season entirely.

Data problem: The useful data are already present but disconnected: historical title-level sales, retailer order history, catalog launch timing, event calendars, promotions, returns, and supplier lead times. Without a forecasting layer, buying decisions rely too heavily on experience and average trends, which is risky when one title can be evergreen and another can have a brief seasonal spike.

AI solution: Innovacio would build a SKU-and-channel demand model using historical sales, title attributes, customer segment, seasonality, and promotion timing. In retail research, a tree-based ensemble model trained on over six years of daily demand data from more than 330 products and 5.2 million records outperformed LSTM across multiple accuracy measures, and the gains were especially strong for more volatile product groups. Another seasonal-items study found that stacked LSTM outperformed SARIMA, Prophet, CNN, and simpler baselines for sales forecasting on real retail data.

What Innovacio would build: a forecasting dashboard that ranks titles by forecast confidence, highlights seasonal uplift windows, and separates evergreen demand from event-driven demand. Buyers would see not just a single forecast number, but a range, a confidence score, and an explanation of which factors are driving the next season's order plan.

Why it matters: Better forecasting reduces both missed seasonal sales and excess inventory. In this kind of business, that means higher fill rate on retailer orders, lower markdown risk, and fewer titles crossing into noncurrent classification.

Evidence from literature	What it means for this case
Tree-based retail demand forecasting on 5.2M records outperformed LSTM across MAPE, MAE, RMSE, and R ² , with more than a 25% MAPE reduction on one volatile product group.	AI can handle high-SKU, mixed-seasonality retail demand better than a single human heuristic.
Seasonal-item forecasting research showed stacked LSTM outperforming SARIMA, Prophet, and CNN on retail sales data.	The business should use seasonal-aware models rather than plain historical averages.

Suggested implementation details

- Forecast at the level that matters operationally: title, channel, retailer group, and season.
- Separate evergreen titles from seasonal and promotional titles because they follow different demand rules.
- Keep a forecast confidence score in the dashboard so buyers know where human review is needed.
- Refresh forecasts before each peak cycle, not just at year-end.

Supplier lead-time and purchase-order planning

Business problem: The annual report says most products are printed overseas and typically require four to eight months of lead time. That is a long planning horizon for a business with rapidly changing retail demand. Supplier minimums also force occasional overbuying, which can create noncurrent inventory even when the sales team is being careful.

Data problem: Procurement needs a better view of supplier lead time, production constraints, order quantities, freight timing, and the probability that a title will still be in demand when it arrives. In a fragmented operation, this information is often spread across email, spreadsheets, ERP screens, and the instincts of the purchasing team.

AI solution: Innovacio would create a procurement cockpit that blends lead-time prediction, supplier performance scoring, and purchase-order recommendations. Instead of only showing what is already late, the model would estimate which print orders should be



accelerated, which suppliers are drifting, and which purchase orders should be reduced because sell-through is weakening.

What Innovacio would build: a dashboard that links each planned PO to forecast demand, expected arrival date, minimum order constraints, and the risk that the title becomes noncurrent before sell-through. Buyers would be able to sort by urgency, margin exposure, and supplier reliability.

Why it matters: The objective is not just faster buying. It is better buying. AI reduces the chance that minimum-order rules create excess stock and helps procurement choose which titles deserve inventory investment before the seasonal window closes.

Procurement control point	AI contribution	Business outcome
Lead-time variability	Predict late or unstable suppliers	Fewer missed seasonal windows
MOQ pressure	Recommend order sizes by demand risk	Less forced overbuying
Import and transit risk	Track inventory in transit against demand windows	Better in-stock position on arrival
Supplier performance	Score vendors by timeliness and reliability	More disciplined sourcing decisions

Retailer replenishment and consignment visibility



Business problem: The company sells to a large number of retail outlets and also uses consignment-style inventory with eligible partners. The report says approximately 17.3% of active brand partners held consignment inventory at the end of fiscal 2025. That creates a visibility problem: the business must know not only what has been shipped, but what has sold, what remains on hand, and what should be replenished or pulled back.

Data problem: Retailer-level sell-through, return rates, promotional performance, and partner-specific order histories are often delayed or inconsistent. Without a unified view, the business risks sending the wrong titles to the wrong stores, or holding too much consignment stock in weak locations.

AI solution: Innovacio would build a retailer operations dashboard with partner scorecards, consignment aging, sell-through trends, and next-best-order suggestions. The system would identify which outlets can absorb more inventory, which ones need a slower replenishment cadence, and which titles should be moved from push to pull mode.

What Innovacio would build: a live view of retailer performance by outlet type, product category, and season, including consignment stock on hand, returns, and projected reorder date. The purchasing and sales teams would use the same screen, reducing mismatch between what is sold in the field and what is ordered at headquarters.

Why it matters: Retailers get the right titles sooner, sales reps can present with better stock confidence, and the business reduces the expensive fog around consignment inventory. That directly supports service levels and cash conversion.

Noncurrent inventory, obsolescence, and returns control

Business problem: The company's filings show noncurrent inventory is a real issue, driven by minimum order requirements and reduced sales volumes. Inventory in excess of 2.5 years of anticipated sales is classified as noncurrent, and the annual report shows more than \$15 million of noncurrent inventory net in 2025. That is a major working-capital risk in a title-based business where some products age better than others.

Data problem: Slow-moving stock is easy to notice too late and hard to manage proactively. Teams need to know not only what is slow today, but what is likely to become slow in the next cycle. That requires aging curves, title lifecycle analysis, and retailer sell-through patterns, not just total on-hand quantity.

AI solution: Innovacio would use inventory-aging models that classify titles into keep, rebalance, discount, or stop-reprint categories. The dashboard would flag titles nearing noncurrent



thresholds, estimate future sell-through, and recommend whether the stock should be allocated to stronger outlets, promoted, bundled, or stopped.

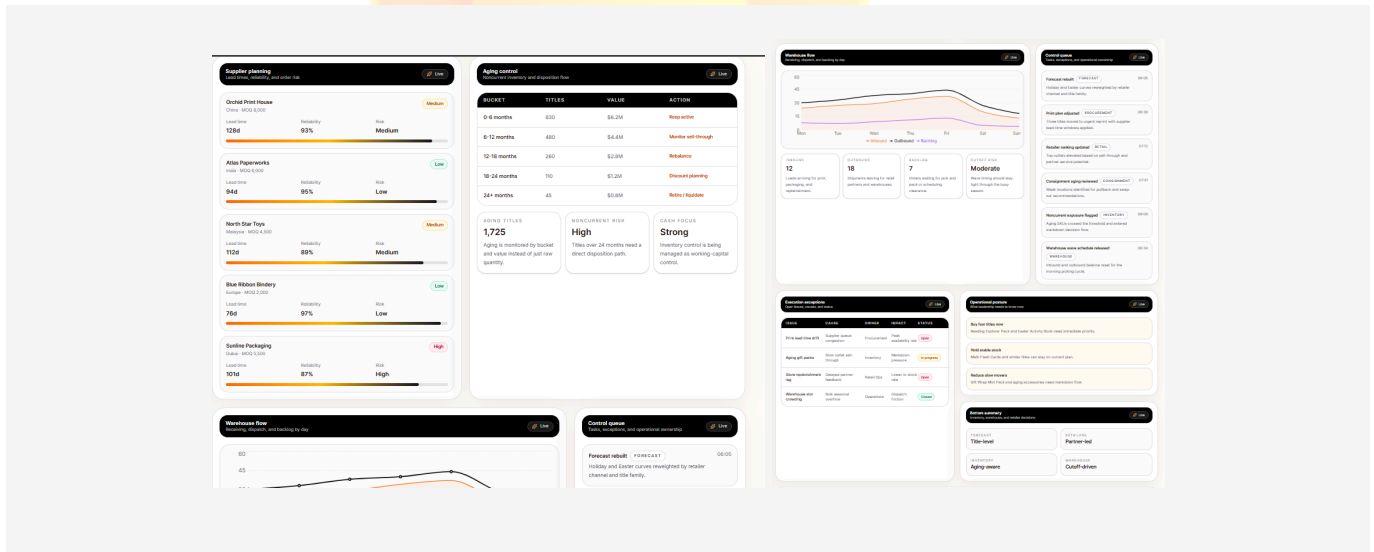
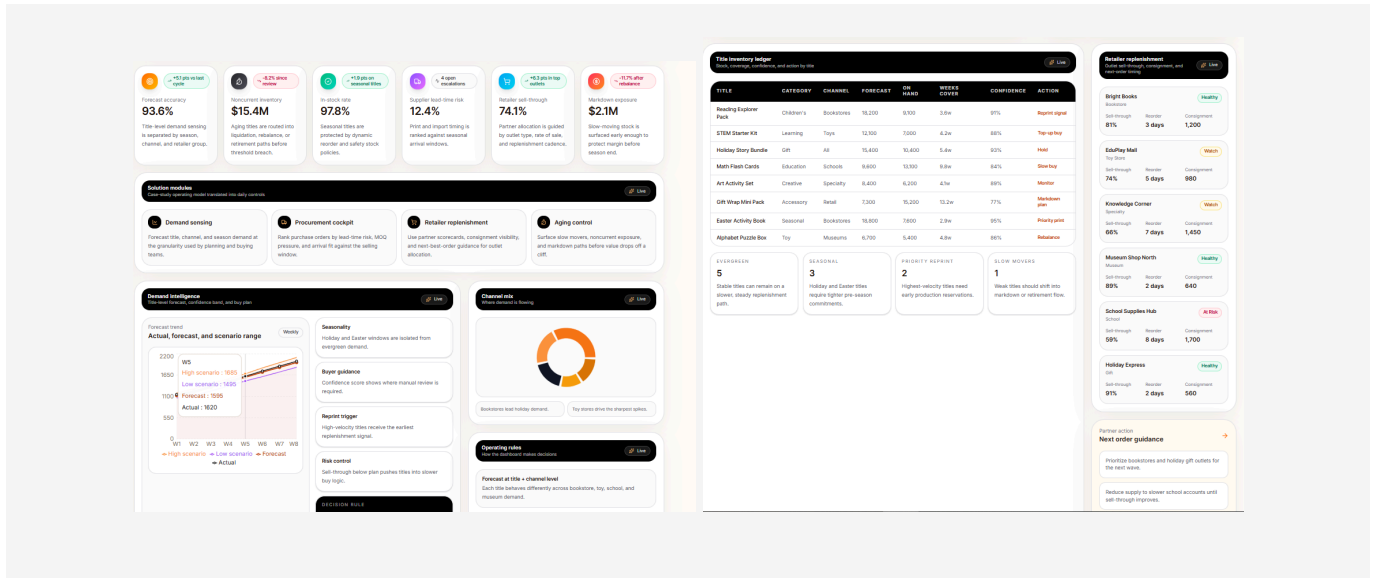
What Innovacio would build: a slow-mover control tower that displays aging by title, channel, and warehouse, with alerts for items drifting toward noncurrent status. The same view would show which titles still have enough demand to justify replenishment and which should be retired from future print planning.

Why it matters: This turns the inventory problem from a year-end accounting issue into a daily operational control problem. The business can protect margin, reduce write-down pressure, and improve warehouse space utilization.

Inventory issue	AI response	Expected operational effect
Slow-moving titles	Aging and sell-through prediction	Lower obsolescence risk
Noncurrent stock	Rebalance and markdown recommendations	Better cash conversion
Returns and cancellations	Return-rate monitoring by retailer type	More accurate replenishment
Warehouse clutter	Multi-echelon stock prioritization	Cleaner inventory posture



Screenshots



Final Takeaway



This business is a strong AI supply-chain case because the operating challenge is not a lack of product ideas; it is orchestration. The company has a wide title catalog, long overseas lead times, seasonal spikes, retailer complexity, and significant inventory aging exposure. Innovacio solves that by turning disconnected data into one operating system for forecasting, procurement, retailer replenishment, and inventory control.

The best story to tell is simple: before AI, the business was forced to guess early and react late. After Innovacio, teams can see demand earlier, buy more intelligently, manage retailer stock with more confidence, and keep noncurrent inventory from silently eroding margin.

Research references used for the case study

- Educational Development Corporation, 2025 Annual Report (Form 10-K). Business description, seasonality, title count, lead times, inventory classification, and consignment inventory disclosures.
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- Ensafi et al. (2022). Time-series forecasting of seasonal items sales using machine learning – A comparative analysis.
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- Rossi (2021). Demand Forecasting. In Predictive Analytics for Inventory Control.

Appendix:

KPI	Why it matters in this business	Good evidence source
Forecast MAPE / RMSE	Shows whether title-level demand is getting more accurate	Forecast engine output
In-stock rate	Measures retailer service level	Retailer / ERP data
Noncurrent inventory value	Shows aging and working-capital pressure	Inventory valuation
Lead-time variance	Measures supply reliability	Supplier and shipment records

KPI	Why it matters in this business	Good evidence source
Consignment sell-through	Shows partner inventory health	Partner inventory reports
Return rate	Shows product and assortment quality	Retail returns and claims

